

Python for Data Science - 2305CS303

Lab - 7

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01) WAP to find their union, intersection, and difference of the given two sets.

Sample Input:

```
A = {1, 2, 3, 4}
B = {3, 4, 5, 6}
```

Sample Output:

```
Union: {1, 2, 3, 4, 5, 6}
Intersection: {3, 4}
Difference (A-B): {1, 2}
```

```
In [8]: A = {1, 2, 3, 4}
        B = {3, 4, 5, 6}

        print("union " ,(A | B))
        print("intersaction " ,(A & B))
        print("difference" ,(A - B))
```

```
union {1, 2, 3, 4, 5, 6}
intersaction {3, 4}
difference {1, 2}
```

02) WAP to Find elements common to all three given sets.

Sample Input:

```
A = {1, 2, 3, 4}
B = {2, 3, 5}
C = {2, 3, 6}
```

Sample Output:

```
{2, 3}
```

```
In [11]: A = {1, 2, 3, 4}
          B = {2, 3, 5}
          C = {2, 3, 6}

          print(A.intersection(B,C))
```

```
{2, 3}
```

03) WAP to remove duplicate elements from a list while preserving unique values.

Sample Input:

```
[10, 20, 10, 30, 20, 40]
```

Sample Output:

```
{10, 20, 30, 40}
```

```
In [19]: lst = [10, 20, 10, 30, 20, 40]
          print(set(lst))
```

```
{40, 10, 20, 30}
```

04) WAP to count how many elements are common between two sets.

Sample Input:

```
A = {1, 2, 3, 4}
B = {3, 4, 5, 6}
```

Sample Output:

```
2
```

```
In [20]: A = {1, 2, 3, 4}
          B = {3, 4, 5, 6}

          print(len(A.intersection(B)))
```

```
2
```

05) WAP to Merge two dictionaries. If a key exists in both dictionaries, add their values.

Sample Input:

```
Dict1 = {1: 10, 2: 20, 3: 30}
Dict2 = {2: 40, 3: 50, 4: 60}
```

Sample Output:

```
{1: 10, 2: 60, 3: 80, 4: 60}
```

```
In [23]: Dict1 = {1: 10, 2: 20, 3: 30}
Dict2 = {2: 40, 3: 50, 4: 60}

mrg = Dict1.copy()

for key ,v in Dict2.items():
    if key in mrg:
        mrg[key]+= v;
    else:
        mrg[key]=v
print(mrg)
```

```
{1: 10, 2: 60, 3: 80, 4: 60}
```

06) WAP to Sort a dictionary in ascending order of its values.

Hint: in sorted(), use udf for key parameter

Sample Input:

```
{'Math': 70, 'Physics': 85, 'Chemistry': 60}
```

Sample Output:

```
{'Chemistry': 60, 'Math': 70, 'Physics': 85}
```

```
In [34]: dic = {'Math': 70, 'Physics': 85, 'Chemistry': 10}

s = dict(sorted(dic.items(),key=lambda x:x[0]))
print(s)
```

```
{'Chemistry': 10, 'Math': 70, 'Physics': 85}
```

07) WAP to find the keys having maximum and minimum values in a dictionary.

Sample Input:

```
{'A': 50, 'B': 90, 'C': 30}
```

Sample Output:

```
Max Value Key: B  
Min Value Key: C
```

```
In [64]: v = {'A': 50, 'B': 90, 'C': 30}

lst = sorted(v.items(),key=lambda x:x[1])
res = list(lst[-1])
print(res[0])
```

B

08) WAP to create a dictionary with numbers from 1 to N as keys and their squares as values.

Sample Input:

```
N = 5
```

Sample Output:

```
{1: 1, 2: 4, 3: 9, 4: 16, 5: 25}
```

```
In [9]: n=5;

s = {x:x**2 for x in range(1,6)}

print(s)
```

```
{1: 1, 2: 4, 3: 9, 4: 16, 5: 25}
```

09) WAP to count the frequency of each character in a string using a dictionary.

Sample Input:

```
programming
```

Sample Output:

```
{'p': 1, 'r': 2, 'o': 1, 'g': 2, 'a': 1, 'm': 2, 'i': 1, 'n': 1}
```

```
In [14]: text = "programming"
dic = {}
for i in text:
    dic[i] = dic.get(i,0)+1

print(dic)
```

```
{'p': 1, 'r': 2, 'o': 1, 'g': 2, 'a': 1, 'm': 2, 'i': 1, 'n': 1}
```

10) WAP to swap keys and values of a dictionary.

Sample Input:

```
{1: 'A', 2: 'B', 3: 'C'}
```

Sample Output:

```
{'A': 1, 'B': 2, 'C': 3}
```

```
In [18]: dic ={1: 'A', 2: 'B', 3: 'C'}
key = {}

for k,v in dic.items():
    key[v]=k

print(key)
```

```
{'A': 1, 'B': 2, 'C': 3}
```

11) WAP to find the student with the highest total marks, given student names and their marks.

Sample Input:

```
{'Amit': [70, 80, 90], 'Riya': [85, 75, 80], 'Neha': [60, 70, 65]}
```

Sample Output:

```
Topper: Amit
```

```
In [39]: dic = {'Amit': [70, 80, 90], 'Riya': [85, 75, 80], 'Neha': [60, 70, 65]}
maxtotal = 0
total=0
topper = ''
for k,v in dic.items():

    total = sum(v)
    if total > maxtotal:
        maxtotal = total
        topper = k

print(topper)
```

```
Amit
```

12) WAP to extract keys whose values are even numbers.

Sample Input:

```
{1: 11, 2: 20, 3: 15, 4: 40}
```

Sample Output:

```
[2, 4]
```

```
In [42]: d = {1: 11, 2: 20, 3: 15, 4: 40}
lst=[]
for k,v in d.items():
    if v%2==0:
        lst.append(k)

print(lst)
```

```
[2, 4]
```

13) WAP to Find common keys between two dictionaries.

Sample Input:

```
Dict1 = {1: 'A', 2: 'B', 3: 'C'}
Dict2 = {2: 'X', 3: 'Y', 4: 'Z'}
```

Sample Output:

```
[2, 3]
```

```
In [44]: Dict1 = {1: 'A', 2: 'B', 3: 'C'}
Dict2 = {2: 'X', 3: 'Y', 4: 'Z'}
lst=[]
for k,v in Dict1.items():
    if k in Dict2:
        lst.append(k)

print(lst)
```

```
[2, 3]
```

14) WAP to handle missing keys in dictionaries.

Sample Input - 1:

```
dict1 = {'a': 5, 'c': 8, 'e': 2}
key = 'c'
```

Sample Output - 1:

```
8
```

Sample Input - 2:

```
dict1 = {'a': 5, 'c': 8, 'e': 2}
key = 'd'
```

Sample Output - 2:

Key Not Found

```
In [ ]: dict1 = {'a': 5, 'c': 8, 'e': 2}
key = 'd'

result = dict1.get(key, "Key Not Found")
print(result)
```

15) WAP to create a dictionary using dictionary comprehension from two lists containing student names and marks. Include only those students in the dictionary whose marks are greater than or equal to 75.

Sample Input:

```
students = ['Amit', 'Riya', 'Neha', 'Karan']
marks = [78, 72, 85, 60]
```

Sample Output:

```
{'Amit': 78, 'Neha': 85}
```

```
In [53]: students = ['Amit', 'Riya', 'Neha', 'Karan']
marks = [78, 72, 85, 60]

dic = dict(zip(students,marks))
res = {}
for k,v in dic.items():
    if v > 75:
        res[k]=v

print(res)
```

```
{'Amit': 78, 'Neha': 85}
```

```
In [ ]:
```